

Review of Clio's cylindric M-convexity notes (UIDs 288/289/291): §5 of note 1 is complete

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Reviewed: (1) 2026-09-20-c1-cylindric-M-convexity.pdf (UID 291 build, supersedes 288/289);
(2) 2026-09-21-c1-affine-stanley-exchange.pdf; (3) 2026-09-21-c2-affine-stanley-321-avoiding.pdf
(UID 291 build).

WIP commit: 41bbe22

Code: projects/work-in-progress/notes/code/2026-09-25-clio288-greedy-check.py

Grades: hunch < sketched < computed < checked-sober < proved.

Verdict

§5 of note 1: COMPLETE. I read it line by line as someone who did not write it. Every step checks. It has no circularity and no missing case. It uses only Lemma 3.1, Lemma 3.2 and Corollary 3.4, and I re-derived all three. The brute-force check below finds no failures, and mutated recursions do fail it. My only remarks are cosmetic additions. There is one real but small gap, in §7/§10 (SNP and the Newton polytope), not in §5.

1 §5 line by line

Def. 5.1 / Lemma 5.2(1). Taking the coordinatewise minimum of two strictly increasing sequences with $x_{i+m} = x_i + n$ gives such a sequence. The induction $g^{t-1} \subseteq \lambda \Rightarrow g^{t-1} \ll g^t, g^t \subseteq \lambda$ is correct. The base case uses $\mu \subseteq \lambda$, which is part of the definition of a skew shape (Rem. 2.4). Say this explicitly. Also, (1) does *not* need $T_\ell \neq \emptyset$; only (2) does. **Proved.**

Lemma 5.2(3). $x_i^t \leq x_{i+1}^{t-1} - 1 \leq g_{i+1}^{t-1} - 1$ and $x_i^t \leq \lambda_i$ (from $x^t \subseteq x^\ell$ by transitivity). This is correct. It uses only half of (1) for $\nu = x^{t-1}, \rho = x^t$. **Proved.**

Lemma 5.2(2). Uses (3) but not the reverse, so there is no circularity. **Proved.**

Prop. 5.4. Correct. The hedge “finite whenever $T_\ell \neq \emptyset$ for some ℓ ” is vacuous (see Remark A): $\ell_0 < \infty$ *always*. **Proved.**

Prop. 5.5. $\hat{\lambda} \in P(W_\ell)$ via Cor. 3.4. For every ν placed in position order, the prefix sums are $\leq$ those of γ (telescoping plus 5.2(3), using $u(x^0) = u(g^0)$), which are in turn $\leq$ those of $\hat{\lambda}$. Correct. **Proved.**

Dependencies outside §5: Cor. 3.4 needs $L_{i+m} + U_{i+m} = L_i + U_i + 2n$ so that the reflected row is still a periodic shape. This holds, but it is not written; add half a line. Lemma 3.2 is correct (I redid the wrap-around bookkeeping). Lemma 4.1 and Prop. 4.2 are correct; $b \leq \ell$ holds because $\sigma_j > 0$. The §6 polymatroid argument is correct.

2 Two free strengthenings you should state

Remark A (closed form; proved). By induction, $g_i^t = \min(\lambda_i, \mu_{i+t} - t)$, because $\lambda_{i+1} - 1 \geq \lambda_i$ absorbs all the intermediate terms. Since $\mu_{i+km} - km = \mu_i + k(n - m) \rightarrow \infty$, it follows that $\ell_0 = \min\{t : \mu_{i+t} - t \geq \lambda_i \ \forall i\} < \infty$ for every $\mu \subseteq \lambda$. So $T_\ell \neq \emptyset \iff \ell \geq \ell_0$ holds unconditionally. The $\ell_{\min} = \infty$ branch of note 3, Lemma 4.1, and its use of $\tilde{F}_w \neq 0$ in Thm. 6.1 are therefore dead weight. *Computed:* 5,947,200 (t, i) -checks, $n \leq 7$, 0 failures.

Remark B (γ is already sorted; proved). Apply 5.5's prefix inequality to $\nu = \hat{\lambda}$ itself, which lies in W_ℓ in position order by Cor. 3.4. This gives $\Lambda_r \leq \gamma_1 + \dots + \gamma_r \leq \Lambda_r$ for all r , so $\gamma = \hat{\lambda}$. The greedy weight is weakly decreasing and “sort” can be dropped. *Computed:* 12,866/12,866.

3 Computation (independent code; mine, not yours)

Chains are enumerated from the raw condition (1), not via Lemma 3.1. Range: $n \leq 7$, all $m < n$, $\mu_1 = 0$, $d \leq 9$, $\ell \leq 6$. This gives 3,032 shapes and 12,866 nonempty $(\lambda/\mu, \ell)$ instances. Checks: 5.2(1), $\gamma \in W_\ell$, the prefix domination in position order, $\hat{\lambda}$ as dominance maximum, $T_\ell \neq \emptyset \iff \ell \geq \ell_0$, ℓ -independence of $\hat{\lambda}$, full support equality of Thm. 7.1, and symmetry of the coefficients. **0 failures.** Negative controls: dropping the -1 in (6) gives 1,250 failures; replacing $g_{i+1}^{t-1} - 1$ by $g_i^{t-1} + 1$ gives 1,058 failures. Anchor: bead model vs. ordinary skew Kostka numbers at $n = 40$, $m \in \{2, 3\}$, $\ell = 4$. This gives 78,823 coefficients with 0 mismatches.

4 The one real gap (not in §5)

Thm. 7.1 asserts SNP and $\text{Newton} = P_{\hat{\lambda}}$. Gap 2 says both are independent of Rado. For SNP that is true, but only through a line you do not write: a lattice point of $\text{conv}(W_\ell)$ satisfies the linear inequalities $\alpha(S) \leq \Lambda_{|S|}$ and $\alpha([\ell]) = d$, so it lies in W_ℓ by the display in Prop. 6.2. Write that line. $\text{conv}(W_\ell) = P_{\hat{\lambda}}$ needs $W_\ell \subseteq P_{\hat{\lambda}}$, and that is Rado's direction. It follows in one line from your own Lemma 4.1: $\nu - e_a + e_b$ lies on the segment from ν to $(ab)\nu$. Add it and gap 2 closes. Grade now: SNP **checked-sober**; Newton identification **sketched**. Both become **proved** after two added lines.

5 Notes 2 and 3: skim, nothing breaks

- Note 2, Prop. 5.7 (13-point counterexample): I recomputed it. Property (2) holds and W is not M-convex (witness $(4, 1, 1), (3, 0, 3), i = 2$). **Computed.** Lemma 4.2, Lemma 4.3 and Cor. 4.5 (the Ψ duality) read correctly. **Checked-sober** on a skim.
- Note 2, Thm. 5.6 cites “[1, §6]” for “equals $P_{\hat{\lambda}} \cap \mathbb{Z}^r$ ”. §6 does not prove that (see §4 above).
- Note 2, §1.3 calls (H4) an “omission” of note 1. In note 1 it is Lemma 5.2(3) and it is proved. Reword this, or a reader will think note 1 has a hole.

- Note 3, Lemma 4.1 now duplicates note 1, Prop. 5.4. Its $\ell_{\min} = \infty$ case is vacuous (Remark A). Cor. 6.3 and Step 4 are correct. The route rests on Lam ~~thm~~:321, and you flag that you cannot follow 1.1868. So Thm. 6.1 is proved modulo a cited theorem with an open reading question. That is an honest statement.
- Cross-reference slips: note 3 cites “[2, Prop. 2.4]” but it is Prop. 2.2. Note 1’s cover block says “Gaps item (iii) of §8” but the Gaps are in §10, items 1–3. The UID 289 email says “Corollary 3.3” but the PDF has Cor. 3.4.

Bottom line

§5 is complete and **proved**. Thm. 7.1 (support + M-convexity) is **proved**. Add Remarks A and B, the periodicity half-line in Cor. 3.4, and the two lines for SNP/Newton. After that, note 1 has no gaps that I can find.